Fundamentals of Programming

Final Review

Question 1

Given a string `s` that contains only alphabetic characters and spaces, complete the following C++ function to reverse the order of the words in the string.

```
string reverseWords(string s);
```

Input	**Output**
`Hello World`	`World Hello`
`HCMUS is amazing`	`amazing is HCMUS`

Question 2

Define a struct named `Circle` to represent a **circle** in the 2D Cartesian plane determined by:

- The coordinates of its center (x, y).
- Its radius r.

Then, complete the following function:

```
void printRelation(Circle c1, Circle c2);
```

This function prints the **geometric relationship** between two circles `c1` and `c2`.

Input	**Output**
$c_1(0,0,3)$, $c_2(0,0,3)$	Coincident
$c_1(0,0,3)$, $c_2(6,0,3)$	Externally tangent
$c_1(0,0,5)$, $c_2(3,0,2)$	Internally tangent
$c_1(0,0,5)$, $c_2(4,0,3)$	Intersect
$c_1(0,0,5)$, $c_2(1,1,1)$	One contains the other
$c_1(0,0,2)$, $c_2(10,0,2)$	Separate

Question 3

A **magic square** is a square matrix of size $n \times n$ that satisfies the following conditions:

- The sums of all rows are equal.
- The sums of all columns are equal.
- The sums of the two main diagonals are equal.

You are given a **2D array using pointers** with the number of rows and columns.
Complete the following C++ function:

```cpp
bool isMagicSquare(int** a, int nRows, int nCols);
```

The function returns:

- `true` if the matrix is a magic square.
- `false` otherwise.

For example
The matrix:

$$\begin{bmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{bmatrix}$$

is a **magic square** because it satisfies the following conditions:

- Row sums:
$$8 + 1 + 6 = 3 + 5 + 7 = 4 + 9 + 2 = 15$$
- Column sums:
$$8 + 3 + 4 = 1 + 5 + 9 = 6 + 7 + 2 = 15$$
- Sums of the two main diagonals:
$$8 + 5 + 2 = 6 + 5 + 4 = 15$$

Therefore, the matrix is a magic square with magic constant 15.

Question 4

Given a singly linked list defined as follows:

```
struct Node {
    int data;
    Node* pNext;
};

struct List {
    Node* pHead;
};
```

Complete the following function to **remove all nodes** whose values are **coprime** with a given integer x.

```
void removeCoprimeNodes(List& l, int x);
```

Notes: Two integers are **coprime** if their greatest common divisor (GCD) is equal to 1.

Input	Output
List: $[6 \to 9 \to 10 \to 14]$, $x = 7$	$[14]$
List: $[4 \to 6 \to 8 \to 9]$, $x = 3$	$[6 \to 9]$

Question 5

Given an input file named `input.txt` that contains information of students. Each line has the following format:

```
Name, Math score, Literature score, English score
```

where the fields are separated by commas:

- `Name` is a string (e.g., `Nguyen Van A`).
- Scores are real numbers.

Write a program that:

- Reads data from `input.txt`.
- Computes the GPA (average of the three scores) for each student.
- Writes to `output.txt` the list of students who have the **highest GPA**.

For example

`input.txt`

```
Nguyen Van A,8,7,9
Tran Thi B,9,9,9
Le Van C,9,9,9
```

`output.txt`

```
Tran Thi B,9.0
Le Van C,9.0
```

Note:

- GPA should be printed with one decimal place.
- If multiple students share the highest GPA, print all of them.

Regulations

Please follow these regulations:

- This is an **INDIVIDUAL** assignment, used for bonus points.
- You may use any text editor (MS Word, Google Docs, LaTeX, etc.) or complete the assignment on paper if preferred. However, the final submission must be exported in **PDF format** and uploaded to Moodle.
- After completing assignment, check your submission before and after uploading to Moodle.
- AI tools are **not restricted**; however, students should use them wisely.
- Any plagiarism, any tricks, or any lie will have a 0 point for the course grade.

The end.